Question 1 2 marks 101

Avery has 4 adventure books, 5 mystery books, and 1 comic book.

Determine the number of ways he can arrange all of the books on his shelf if each type of book must be grouped together.

Question 2 4 marks 102

Solve the following equation, algebraically, over the interval $[0, 2\pi]$.

$$5\cos^2\theta - \cos\theta - \sin^2\theta = 0$$

Question 3 3 marks 103

Given that $-6048x^2y^5$ is the sixth term in the expansion of $(3x-m)^7$, determine m.

Question 4 a) 2 marks b) 1 mark 104 105

A series of blood tests measures the concentration of a prescribed drug. This concentration decreases according to the formula $A = Pe^{rt}$ where:

A is the concentration at time t

P is the initial concentration

r is the rate of change

t is the time, in hours, after the first blood test

The initial concentration is 3.8900 units/mL. Three hours later, the concentration is 1.7505 units/mL.

a) Determine the rate of change, r, algebraically.

b) Determine the concentration of the prescribed drug four hours after the initial concentration of 3.8900 units/mL was measured. Express the answer correct to 4 decimal places.

Note: A calculator is not required for the remaining test questions.

Question 5 | 1 mark 106

Ariane uses the formula $s = \theta r$ to determine the arc length of a circle that has a central angle of $20°$ and a radius of 15 cm.

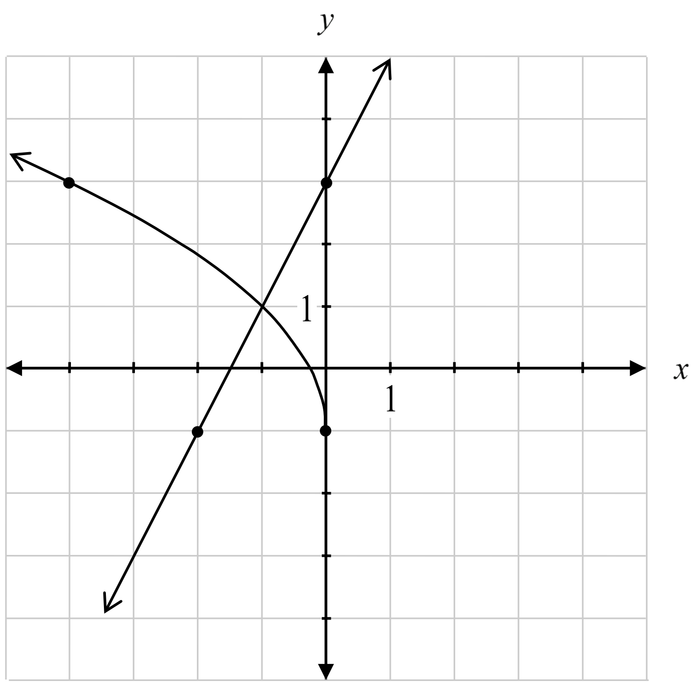
Below is Ariane's work:

$$s = \theta r$$
$$s = (20)(15)$$
$$s = 300 \text{ cm}$$

Describe her error.

Question 6 1 mark 107

Using the graphs below, state the solution of the equation $2x+3=2\sqrt{-x}-1$.



Question 7 3 marks 108

Solve, algebraically.

$$\log_2(x+3) = 5 - \log_2(x-1)$$

Question 8 1 mark 109

Explain why the value of n must be greater than or equal to the value of r, when using ${}_{n}C_{r}$.

Question 9 3 marks 110

Given that $y = |x|$, determine the equation of the resulting function, $g(x)$, after the following transformations:

- reflection in the x-axis
- vertical translation 5 units down
- horizontal stretch by a factor of 3

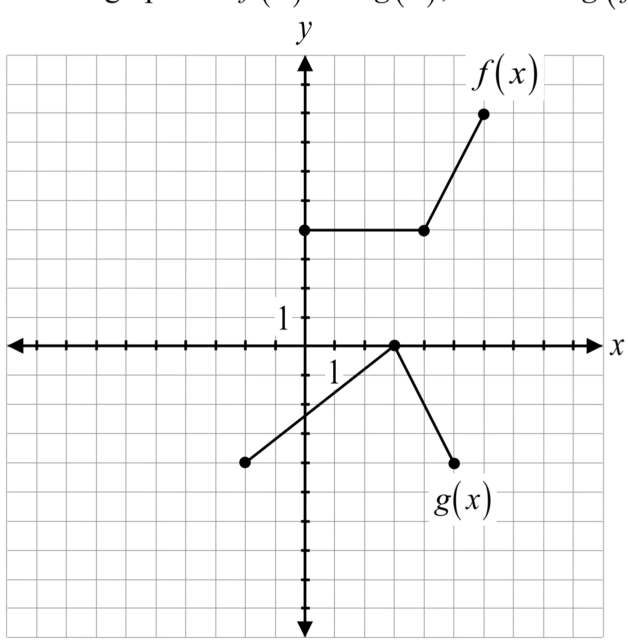
$g(x) =$ ______________________________

Question 10 — 1 mark 111

Explain why the graph of $y = \dfrac{x-1}{x^2 + x - 6}$ has a horizontal asymptote at $y = 0$.

Question 11 — 1 mark 112

Given the graphs of $f(x)$ and $g(x)$, evaluate $g\left(f(2)\right)$.



Question 12 1 mark 113

Kennedy was asked to solve the equation $\tan\theta = 1$ over all real numbers.

Below is Kennedy's solution:

$$\tan\theta = 1$$

$$\theta = \frac{\pi}{4}, \frac{5\pi}{4}$$

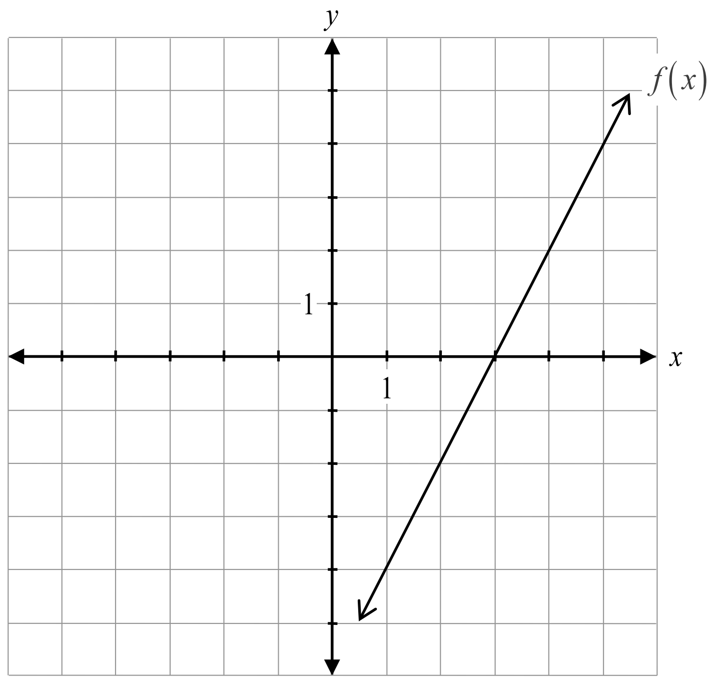
Describe her error.

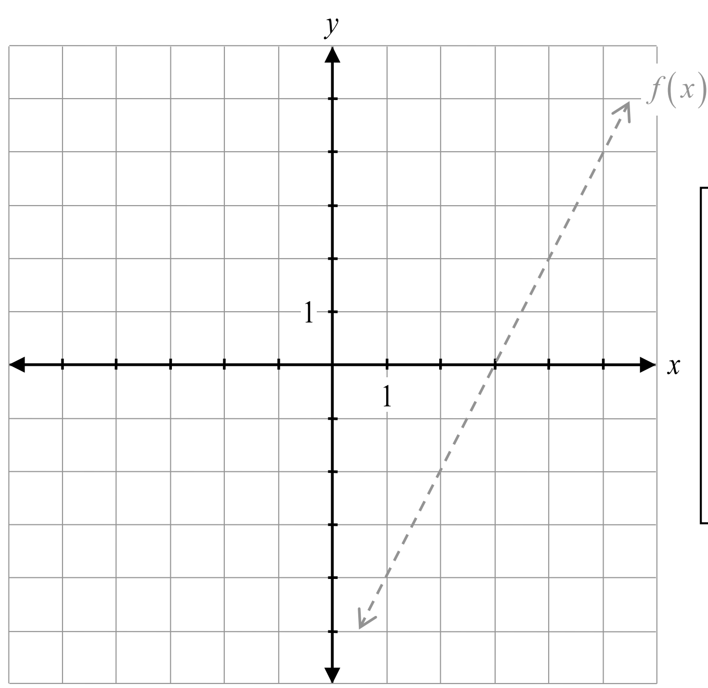
Question 13 3 marks 114

Solve, algebraically.

$$_{n}C_{3} = 3\left({_{n}P_{2}}\right)$$

Given the graph of $y = f(x)$, sketch the graph of $y = \sqrt{f(x)}$.





The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 15 3 marks 116

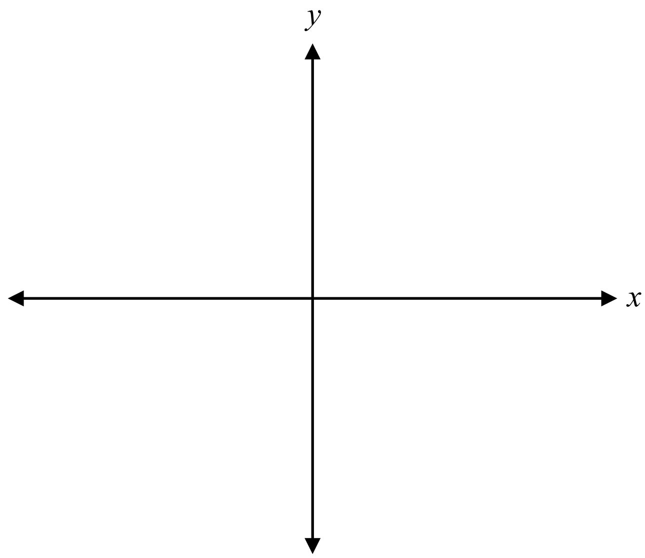
Prove the identity for all permissible values of θ.

$$\frac{\sec\theta - \tan\theta\sin\theta}{\tan\theta\sin\theta} = \csc^2\theta - 1$$

Left-Hand Side	Right-Hand Side

Question 16 1 mark 117

Sketch the angle of $-\frac{\pi}{12}$ radians in standard position.



Question 17 1 mark 118

Given that $h(x) = 2x^2 - 7x - 15$, determine possible equations of the functions $f(x)$ and $g(x)$ if $h(x) = f(x) \bullet g(x)$.

$f(x) =$ ______________________________

$g(x) =$ ______________________________

No marks will be awarded for work done on this page.

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